

Fuzzy Logic Systems

Soft Computing — Week 02 Study Booklet

ENGLISH EDITION · VISION-VERIFIED MATH

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Scope: Fuzzy Logic motivation & architecture · Crisp/Classical Set Theory · Fuzzy Sets & Membership Functions

Math source: Vision-verified booklet pages (W02_Fuzzy_Math_Vision_Verified.md)

Notation: $\varnothing$ = empty set · X / U = universe · $\mu: X \rightarrow [0, 1]$ · membership is **not** probability

Language: English only · equations typeset in LaTeX (MathJax)

Source booklet: W02_Fuzzy_Logic_Systems.pdf (pp. 1–66) · extraction Markdown preserved as-is
Condensed study booklet (~28 pages) · Koko · 2026-09-22

This booklet reconstructs Week 02 as one coherent argument: exact sets first, then gradual membership. Source pages are marked beside formulas. Math symbols follow the booklet's own notation (vision-verified).

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3	Fuzzy Logic System architecture (pipeline)
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Learning outcomes

- Explain why fuzzy logic targets **acceptable** reasoning under imprecision.
- Draw and label the four-stage Fuzzy Logic System pipeline.
- Distinguish fuzzification (input) from defuzzification (output).
- Compute classical set operations and state the standard laws.
- Define a fuzzy set by membership function and ordered pairs.
- Write discrete and continuous fuzzy-set notation without treating / as division.

Core idea (source pp. 2–3)

Fuzzy Logic is designed to produce **acceptable reasoning** under imprecision and uncertainty — not necessarily perfectly exact reasoning. It embodies human-like thinking into a control system and can emulate human deductive thinking (inferring conclusions from what people know).

Why fuzzy systems are built

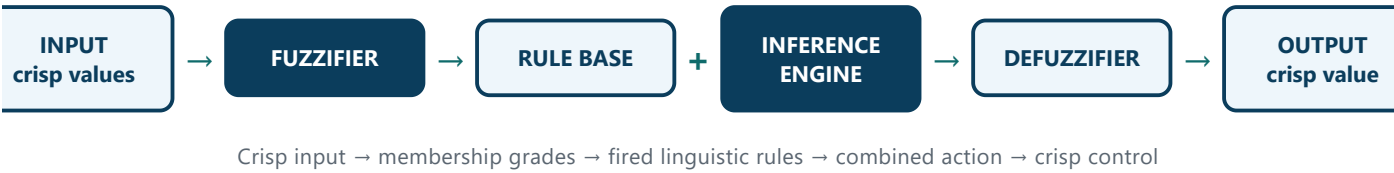
- **Any input quality:** works with imprecise, distorted, or noisy input information.
- **Simple construction:** easy to build and understand; algorithms can be described with little data (low memory).
- **Mathematical base:** grounded in set theory; the reasoning itself is simple.
- **Human-like decisions:** resembles human reasoning and decision making in complex situations.
- **Uncertainty:** uncertainties can be handled with the help of fuzzy logic.

Central progression of Week 02

Real-world imprecision / noise → Human-like linguistic reasoning → Fuzzy Logic System → Fuzzification → Rule Base + Inference → Defuzzification → Crisp control / output value

Mathematical bridge: classical sets define exact membership; fuzzy sets relax membership from $\{0, 1\}$ to the continuous interval $[0, 1]$.

A Fuzzy Logic System converts crisp sensor readings into linguistic grades, reasons with expert rules, then converts the result back into a usable crisp control value (source p. 4 architecture; components pp. 5–6).



Component	Role	Direction
Rule Base	Expert IF–THEN linguistic rules that govern decision making.	knowledge store
Fuzzifier (Fuzzification)	Converts crisp numbers into fuzzy sets / membership degrees.	input conversion
Inference Engine	Matches current fuzzy input to each rule; fires rules; combines control actions.	reasoning
Defuzzifier (Defuzzification)	Converts the inferred fuzzy set into a crisp output value.	output conversion

Exam trap — do not swap directions. Fuzzification is **input** conversion (crisp → fuzzy). Defuzzification is **output** conversion (fuzzy → crisp).

Rule Base (source p. 5)

Contains the set of rules and the IF–THEN conditions provided by **experts** to govern the decision-making system, on the basis of **linguistic information**. Recent fuzzy theory offers methods for design and tuning of fuzzy controllers; most reduce the number of fuzzy rules.

Linguistic IF–THEN form

IF *antecedent* THEN *consequent*

Worked linguistic rule

IF temperature is **high** THEN fan speed is **fast**.
IF pressure is **low** THEN valve is **slightly open**.

Part	Meaning	Typical content
Antecedent	Condition on input linguistic terms	"temperature is high"
Consequent	Action / output linguistic term	"fan speed is fast"
Linguistic variable	Variable taking words as values	temperature, speed, age
Linguistic value	Word / term (fuzzy set label)	high, fast, young, tall

Rules are not Boolean switches. Each rule fires to a **degree** when the input matches its antecedent; several rules can fire together and are combined by the inference engine.

Exam trap. The Rule Base stores **linguistic** expert knowledge (IF–THEN), not raw sensor numbers. Sensors enter through fuzzification.

Fuzzification (source p. 5)

Converts **inputs** — crisp numbers — into **fuzzy sets**. Crisp inputs are the exact values measured by sensors and passed into the control system, such as temperature, pressure, rpm, etc.

$$x \mapsto \mu_{\tilde{A}}(x)$$

(the map from element to membership grade is called fuzzification)

Inference Engine (source p. 6)

Determines the **matching degree** of the current fuzzy input with respect to each rule, decides which rules are to be **fired**, then **combines** the fired rules to form the control actions.

Defuzzification (source p. 6)

Converts the fuzzy sets obtained by the inference engine into a **crisp value**. Several defuzzification methods exist; the best-suited one is used with a specific expert system to reduce error.

Stage	Converts	Example
Fuzzification	crisp → fuzzy	32.5 °C → $\mu_{\text{high}}(32.5) = 0.7$
Inference	fuzzy + rules → fired rules	combine “fan fast” degrees
Defuzzification	fuzzy → crisp	output fan PWM = 180

Exam trap. Fuzzification = **input** path. Defuzzification = **output** path. Never reverse them.

Crisp / classical

- Employs **bi-valued** logic.
- Membership $\mu_A(x) \in \{0, 1\}$.
- Strict boundary: member or non-member.
- Historically tied to Boolean expert systems.

Fuzzy

- Implements **infinite-valued** logic.
- Membership grades in $[0, 1]$.
- Vague boundary; partial belonging.
- Imitates human qualitative thinking.

Scientists argued that human thinking does not always follow crisp “yes/no” logic; it can be vague, qualitative, uncertain, imprecise, or fuzzy. Fuzzy set theory was developed to imitate that style of thinking (source p. 7).

Membership shape (source pp. 8–10, 52)

	Crisp step	Fuzzy curve
“Tall” at height h	$\mu = 0$ below cut (e.g. 5’10”), $\mu = 1$ above	μ rises gradually (e.g. 0.5 ... 0.9 ... 1.0)
Boundary	vertical jump	smooth / gradual
Age words	hard bins	young / middle-aged / old overlap

Tall-people comparison (source p. 52)

Crisp set A: person is tall iff height $\geq 5'10''$ (hard cut).

Fuzzy set $\tilde{A}$: height $5'10'' \rightarrow \mu \approx 0.5$; $6'2'' \rightarrow \mu \approx 0.9$; higher $\rightarrow \mu = 1.0$ (graded curve).

$\mu_A(x) \in \{0, 1\}$ crisp / classical

$\mu_{\tilde{A}}(x) \in [0, 1]$ fuzzy (0 and 1 inclusive)

Grade $\mu(x)$	Meaning
0	no membership — x is not in the set
1	full membership — x is totally in the set
$0 < \mu(x) < 1$	partial membership / degree of belonging

Larger number in $[0, 1] \Rightarrow$ stronger belonging. A fuzzy set is therefore a **vague boundary set** compared with a crisp set (source pp. 55–58).

Definition 1 — Membership function (source p. 55, vision-locked)
If X is a universe of discourse and $x \in X$, then a fuzzy set $\tilde{A}$ in X is a set of ordered pairs:

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}$$

where $\mu_{\tilde{A}}(x)$ maps each element of X onto a membership grade in $[0, 1]$ **both inclusive**.

Exam trap (source p. 48). The degree of membership / truth is **not the same as probability**. Fuzzy truth represents membership in vaguely defined sets — not the chance of an event.

Definition — Crisp / classical set (source p. 11)

A set is an **unordered collection of different elements**. Changing the order of elements or repeating an element does **not** change the set.

Examples from the booklet

set of all positive integers · set of planets in the solar system · set of states in India · set of lowercase letters of the alphabet

Roster / list notation (source p. 12)

List all elements in braces, separated by commas.

$$A = \{a, e, i, o, u\}$$

$$B = \{1, 3, 5, 7, 9\}$$

A = vowels in the English alphabet · B = odd numbers less than 10.

Set-builder notation (source p. 13)

Define the set by a common property $p(x)$:

$$A = \{x : p(x)\}$$

$$A = \{x : x \text{ is a vowel in English}\}$$

$$B = \{x : 1 \leq x < 10 \text{ and } (x \bmod 2) \neq 0\}$$

Member and non-member (source p. 14)

$x \in S$ x is a member of S

$y \notin S$ y is not a member of S

Example If $S = \{1, 1.2, 1.7, 2\}$, then $1 \in S$ but $1.5 \notin S$.

Cardinality (source pp. 15–17)

Cardinality of a set S , written $|S|$, is the number of elements of the set (the cardinal number). An infinite set has infinite cardinality.

$|\{1, 4, 3, 5\}| = 4$

$|\{1, 2, 3, 4, 5, \dots\}| = \infty$

Relation	Meaning	Mapping
$ X = Y $	same cardinality	there exists a bijective $f : X \rightarrow Y$
$ X \leq Y $	X not larger than Y	there exists an injective $f : X \rightarrow Y$
$ X < Y $	strictly smaller	injective but not bijective

Exam trap. Cardinality counts **distinct** elements. Duplicates and order never change a classical set.

Type	Definition	Booklet example
Universal set	All elements in a particular context / application; every set in that context is a subset of it. Written U .	U = all animals on earth; mammals, fishes, insects $\subseteq U$
Finite set	Contains a definite number of elements.	$S = \{x \mid x \in \mathbb{N}, 50 < x < 70\}$
Infinite set	Contains infinitely many elements.	$S = \{x \mid x \in \mathbb{N}, x > 10\}$
Subset	$Y \subseteq X$ if every element of Y is in X .	$Y = \{1, 2\}, X = \{1, 2, 3, 4, 5, 6\} \Rightarrow Y \subseteq X$
Proper subset	Subset of but not equal to: $Y \subset X$ and $ Y < X $.	same $Y \subset X$; X has extra elements

Note: the booklet writes subset with a “less-than-or-equal / contains” style glyph in places; read as $Y \subseteq X$ (every element of Y is in X).

Equal vs subset If $X = Y = \{1, 2, 3\}$, then $Y \subseteq X$ but Y is **not** a proper subset of X .

Type	Definition	Booklet example
Empty / null set	Contains no elements. Written φ in this booklet. Finite; $ \varphi = 0$.	$S = \{x \mid x \in \mathbb{N}, 7 < x < 8\} = \varphi$
Singleton / unit set	Contains exactly one element. Written $\{s\}$.	$S = \{x \mid x \in \mathbb{N}, 7 < x < 9\} = \{8\}$
Equal sets	Contain the same elements (order irrelevant).	$A = \{1, 2, 6\}, B = \{6, 1, 2\} \Rightarrow A = B$
Equivalent sets	Same cardinality, not necessarily same elements.	$A = \{1, 2, 6\}, B = \{16, 17, 22\},$ $ A = B = 3$
Overlapping sets	Share at least one common element.	$A = \{1, 2, 6\}, B = \{6, 12, 42\}$ share 6
Disjoint sets	No common element: $A \cap B = \varphi$.	$A = \{1, 2, 6\}, B = \{7, 9, 14\}$

$$A \cap B = \varphi \quad (\text{disjoint})$$

Exam trap. **Equal** $\Rightarrow$ same elements. **Equivalent** $\Rightarrow$ same size only. Overlapping needs ≥ 1 shared element; disjoint needs 0.

Set operations: union, intersection, difference, complement, Cartesian product (source p. 25).

Union (pp. 25–26)

$$A \cup B = \{x \mid x \in A \text{ OR } x \in B\}$$

Example $A = \{10, 11, 12, 13\}$, $B = \{13, 14, 15\}$
 $A \cup B = \{10, 11, 12, 13, 14, 15\}$ — the common element occurs **only once**.

Intersection (pp. 27–28)

$$A \cap B = \{x \mid x \in A \text{ AND } x \in B\}$$

Same sets $A \cap B = \{13\}$ — only the common element is kept.

$$\begin{aligned} A &= \{10, 11, 12, \mathbf{13}\} & B &= \{\mathbf{13}, 14, 15\} \\ A \cup B &= \{10, 11, 12, 13, 14, 15\} & | & \quad A \cap B = \{13\} \end{aligned}$$

Difference / relative complement (pp. 29–30)

$$A - B = \{x \mid x \in A \text{ AND } x \notin B\}$$

Example (booklet) $A = \{10, 11, 12, 13\}$, $B = \{13, 14, 15\}$
 $(A - B) = \{10, 11, 12\}$ $(B - A) = \{14, 15\}$
Hence $(A - B) \neq (B - A)$ — difference is **not commutative**.

Complement of a set (pp. 31–32)

$$A' = \{x \mid x \notin A\} = U - A$$

More specifically $A' = U - A$, where U is the universal set. The booklet also writes complement with an overline: $\bar{A}$ or $\overline{A}$.

Example If $A = \{x : x \text{ is an odd integer}\}$, then $A' = \{y : y \text{ is not an odd integer}\}$.

Exam trap. $A - B$ keeps what is only in A. It is directional: $A - B \neq B - A$ in general.

Cartesian product / cross product (source p. 33)

For sets $A_1, A_2, \dots, A_n$ the Cartesian product $A_1 \times A_2 \times \dots \times A_n$ is all possible **ordered** n -tuples $(x_1, x_2, \dots, x_n)$ with $x_i \in A_i$.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Booklet example $A = \{a, b\}, B = \{1, 2\}$
 $A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$
 $B \times A = \{(1, a), (1, b), (2, a), (2, b)\}$

Exam trap. Pairs are **ordered**: $A \times B \neq B \times A$ in general. $(a, 1) \neq (1, a)$.

Ordered pairs and the fuzzy-set representation

A fuzzy set $\tilde{A}$ on universe X is written as a set of ordered pairs $(x, \mu_{\tilde{A}}(x))$. Read that precisely: the set is the **graph of the membership function**, and it is a **subset of the Cartesian product** $X \times [0, 1]$.

What the product supplies	The ambient space of all conceivable (element, grade) pairs — necessary, but not sufficient.
What actually selects the pairs	The membership function $\mu_{\tilde{A}}$, which fixes exactly one grade for each $x \in X$.
Therefore	The fuzzy set is determined by its membership function , not by the Cartesian product.

Precision trap. $X \times [0, 1]$ is the space the fuzzy set **lives in**, not the fuzzy set itself, and an arbitrary subset of it need not be one. A fuzzy set on X is a **function** $X \rightarrow [0, 1]$, so its graph carries exactly one grade per element.

Source note: page 33 defines the Cartesian product and gives the $\{a, b\} \times \{1, 2\}$ example only. The link to fuzzy representation is an **author's bridge**, not a claim made on that page.

Vision-verified from booklet pages 34–36. Symbols: $\varnothing$ = empty set · X = universal set (Identity slide).

Commutative (p. 34)

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative (p. 35)

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive (p. 36)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Reading note — Commutative: order of two sets does not matter. Associative: grouping of three does not matter. Distributive: $\cup$ and $\cap$ distribute over each other (both directions).

Exam trap. There is **no** “cancellation law” for $\cup$ / $\cap$ like in ordinary algebra. Do not invent $A \cup B = A \cup C \Rightarrow B = C$.

Vision-verified from booklet pages 37–39.

Idempotency (p. 37)

$$A \cup A = A$$

$$A \cap A = A$$

Identity — four laws (p. 38)

$$A \cup \varnothing = A$$

$$A \cap X = A$$

$$A \cap \varnothing = \varnothing$$

$$A \cup X = X$$

Transitive (p. 39)

$$A \subseteq B \subseteq C \Rightarrow A \subseteq C$$

Involution (p. 39)

$$\overline{\overline{A}} = A$$

Identity slide text. “For set A and universal set X , this property states” the four laws above. Booklet empty set is written $\varnothing$.

Exam trap. Identity has **four** laws (not two). Empty = $\varnothing$; universal on this slide = X (elsewhere U also appears).

De Morgan’s Law (source p. 40, vision-verified)
"It is a very important law and supports in proving tautologies and contradiction."

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

The complement bar on the left covers the **whole** expression. Involution form: $\overline{\bar{A}} = A$.

Verification sketch
An element is outside $A \cap B$ iff it is outside A **or** outside B $\Rightarrow \overline{A \cap B} = \bar{A} \cup \bar{B}$.
An element is outside $A \cup B$ iff it is outside A **and** outside B $\Rightarrow \overline{A \cup B} = \bar{A} \cap \bar{B}$.

Law	Formula (booklet style)
De Morgan (intersection)	$\text{complement}(A \cap B) = \bar{A} \cup \bar{B}$
De Morgan (union)	$\text{complement}(A \cup B) = \bar{A} \cap \bar{B}$
Involution	$\overline{\bar{A}} = A$

Active recall from source pp. 41–44. Attempt before looking at the checklist at the end.

Home Task 1

Let $U = \{1, 2, \dots, 20\}$.

$A = \{2, 5, 7, 8, 10, 13, 15, 16, 18, 19, 20\}$

$B = \{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 14, 16, 17, 19\}$

Find: (1) $A \cup B$ (2) $A \cap B$ (3) $A - B$ (4) $B - A$ (5) A' (6) B'

Home Task 2

Let $A = \{R, S\}$, $B = \{7, 10, 18\}$. Find (1) $A \times B$ (2) $B \times A$.

Home Task 3

Give one example each of: Empty Set · Unit Set · Equal Set · Equivalent Set · Overlapping Set · Disjoint Set.

Self-check (Task 1). $A \cup B = U$. $A \cap B = \{2, 5, 10, 16, 19\}$. $A - B = \{7, 8, 13, 15, 18, 20\}$.
 $B - A = \{1, 3, 4, 6, 9, 11, 12, 14, 17\}$. $A' = U - A$, $B' = U - B$. Verify $A - B \neq B - A$.

Concept of a fuzzy system (source p. 45)

Fuzzy element(s) → Fuzzy set(s) → Fuzzy rule(s) → Fuzzy implication (inference) → Fuzzy system → OUTPUT

Fuzzy sets extend classical sets to handle **partial membership**: elements may belong with varying degrees (source p. 46).

Where crisp boundaries break

Linguistic concept	Crisp cut problem	Fuzzy response
"Tall person"	5'10" tall = tall, 5'09.9" = not tall	gradual $\mu(h)$ curve
"Young / old"	hard age bins, sharp walls	overlapping age terms
"Comfortable city"	city is or is not comfortable	grades like 0.7, 0.9, 0.3
Noisy sensor	single threshold thrashing	stable degrees of belonging

Booklet comfort example (p. 59)

X = all cities in India; A = "city of comfort"

$A = \{(\text{New Delhi}, 0.7), (\text{Bangalore}, 0.9), (\text{Chennai}, 0.8), (\text{Hyderabad}, 0.6), (\text{Kolkata}, 0.3), (\text{Kharagpur}, 0)\}$

Exam trap. Human vague/qualitative thinking is the motivation; the mathematical tool is graded membership in $[0, 1]$.

Basis	Crisp set	Fuzzy set
Form	$S = \{s \mid s \in X\}$ — collection of elements	$F = \{(s, \mu) \mid s \in X\}$ — ordered pairs; $\mu(s)$ is the degree of s
Inclusion of $s \in X$	crisp: strict boundary, yes or no	fuzzy: if present, with a degree of membership
Basic property	precise and certain characteristics	vague or ambiguous properties
Partial membership	not allowed	allowed; element may be partly in the set
Logic	bi-valued	infinite-valued
Typical application	digital design	fuzzy controllers
Truth values	True / False $\in \{0, 1\}$	membership values on $[0, 1]$
Boundary	strict boundary T or F	fuzzy boundary with degree of membership
Crisp / fuzzy relation	can be viewed as a special fuzzy case	a fuzzy set need not be crisp
Excluded middle & non-contradiction	classical discussion — may or may not hold in the booklet's slide wording	booklet slide states they hold (read with care for exam wording)

Exam trap (source p. 48). Degree of membership / truth $\neq$ probability. Fuzzy truth = membership in a vaguely defined set.

Formal definition (source pp. 55–56, 60)

A fuzzy set $\tilde{A}$ in universe of discourse X is a set of ordered pairs:

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}$$

where $\mu_{\tilde{A}}(x)$ is the **membership function** of $\tilde{A}$, and X is the **universe of discourse**. A fuzzy set is totally characterized by its membership function (MF).

Membership function	$\mu_{\tilde{A}}$ — assigns a grade to each x
Universe of discourse	X (also written U on some slides)
Fuzzy set	$\tilde{A}$ — collection of graded memberships

The membership function associates each element $x \in X$ with a value in $[0, 1]$. Compared with a crisp set, a fuzzy set has a **vague boundary**.

$$\mu_{\tilde{A}} : X \rightarrow [0, 1]$$

$[0, 1]$ means all real numbers between 0 and 1 **including** 0 and 1 (vision-locked p. 55).

Case	Value	Meaning (source p. 58)
total membership	$\mu_{\tilde{A}}(x) = 1$	x is totally in $\tilde{A}$
no membership	$\mu_{\tilde{A}}(x) = 0$	x is not in $\tilde{A}$
partial membership	$0 < \mu_{\tilde{A}}(x) < 1$	x is partly in $\tilde{A}$

Universe of discourse

X is the set of all possible values for the variable of interest (height, age, temperature, city names). The membership function “cuts” this universe into graded belonging rather than two boxes.

Fuzzification (source p. 57)

The translation $x \mapsto \mu_{\tilde{A}}(x)$ is known as **Fuzzification**.

Meaning of the membership grade (source p. 57)

- Value of $\mu(x)$ lies between 0 and 1.
 - It is the **degree of membership** (membership value) of element x in set A .
 - Members of a fuzzy set are members **to some degree** — membership grade / degree of membership.
 - The grade is the degree of **belonging**. The larger the number in $[0, 1]$, the stronger the belonging.
- N.B. This is not a probability.** (explicit booklet warning, p. 57; reconfirmed p. 48: fuzzy truth represents membership in vaguely defined sets.)

	Membership $\mu(x)$	Probability $P(E)$
Question	How much is x in the vague set?	How likely is event E ?
Sum rule	Σf = union of grades (not a total of 1)	$\Sigma P = 1$ over exhaustive events
Interpretation	compatibility / belonging	chance / frequency

City of comfort (p. 59) Bangalore at 0.9 means Bangalore is highly compatible with the vague concept “comfortable city” — not that Bangalore is comfortable with probability 0.9.

Alternative notation (source pp. 61–62, vision-verified)

If universe X is **discrete**:

$$\tilde{A} = \sum_{x_i \in X} \mu_{\tilde{A}}(x_i) / x_i$$

Equivalently, for members $x_1, \dots, x_n$ with grades $\mu_1, \dots, \mu_n$:

$$\tilde{A} = \mu_1/x_1 + \mu_2/x_2 + \dots + \mu_n/x_n = \sum_{i=1..n} \mu_i/x_i$$

Notation μ/x means: x is a member of the set **to degree** μ .

Critical reading note (blue text on slide p. 61). “ Σ and integral signs stand for the **union of membership grades**; ‘/’ stands for a **marker** and **does not imply division**.”

Micro-example. $X = \{a, b, c\}$, grades 0.2, 1.0, 0.7 $\Rightarrow \tilde{A} = 0.2/a + 1.0/b + 0.7/c$ (markers, not quotients).

Continuous universe (source pp. 61–62)
If universe X is **continuous**:

$$\tilde{A} = \int_X \mu_{\tilde{A}}(x) / x$$

Again: $\int$ collects / unions membership grades over X ; $/$ is a marker, not division.

Universe	Notation	Symbol meaning
Discrete X	$\sum_{x_i \in X} \mu(x_i) / x_i$	$\sum$ or $\int$ = union of membership grades $/$ = marker , not division
Continuous X	$\int_X \mu(x) / x$	

Exam trap. Do not “compute” μ/x as a quotient. Do not interpret $\int \mu/x$ as an ordinary probability integral.

Compact discrete form (p. 63). $\tilde{A} = \sum_{i=1..n} \mu_i / x_i$ with $x_1, \dots, x_n$ members of $\tilde{A}$ and $\mu_1, \dots, \mu_n$ their degrees of membership.

Representation — mathematical concept (source p. 64, vision-locked)

A fuzzy set $\tilde{A}$ in the universe of information U is a set of ordered pairs:

$$\tilde{A} = \{(y, \mu_{\tilde{A}}(y)) \mid y \in U\}$$

$\mu_{\tilde{A}}(y)$ = degree of membership of y in $\tilde{A}$, with $\mu_{\tilde{A}}(y) \in [0, 1]$.

Case 1 — discrete and finite universe (p. 65)

$$\tilde{A} = \mu_{\tilde{A}}(y_1)/y_1 + \mu_{\tilde{A}}(y_2)/y_2 + \mu_{\tilde{A}}(y_3)/y_3 + \cdots = \sum_i \mu_{\tilde{A}}(y_i)/y_i$$

Slash notation μ_i/y_i means “ y_i at grade μ_i ”. The slash is a **marker**, not division (slide p. 61).

Consistent example. $U = \{\text{New Delhi, Bangalore, Chennai}\}$ with grades 0.7, 0.9, 0.8 can be written 0.7/New Delhi + 0.9/Bangalore + 0.8/Chennai.

Case 2 — continuous and infinite universe (p. 66)

When the universe of information U is continuous and infinite, membership is given by a curve $\mu_{\tilde{A}}(y)$ and the set is written:

$$\tilde{A} = \int_U \mu_{\tilde{A}}(y) / y$$

Case	Universe	Representation
1	discrete & finite	$\sum_i \mu_i / y_i$ or list of pairs (y_i, μ_i)
2	continuous & infinite	$\int_U \mu(y) / y$ or MF curve over U

Reading the diagram idea. Discrete: a finite column of (element, grade). Continuous: a smooth membership curve over a real interval (height, temperature, age).

Exam trap. Both cases use the same semantics — graded membership. Only the bookkeeping symbol changes: $\sum$ vs $\int$. Universe letter may be X or U in this booklet.

Formula sheet (booklet style, vision-safe)

Classical laws

Commutative: $A \cup B = B \cup A \cdot A \cap B = B \cap A$ Associative: $A \cup (B \cup C) = (A \cup B) \cup C \cdot A \cap (B \cap C) = (A \cap B) \cap C$ Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \cdot A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ Idempotent: $A \cup A = A \cdot A \cap A = A$ Identity: $A \cup \varphi = A \cdot A \cap X = A \cdot A \cap \varphi = \varphi \cdot A \cup X = X$ Transitive: $A \subseteq B \subseteq C \Rightarrow A \subseteq C$ Involution: $\overline{\overline{A}} = A$ De Morgan: $\overline{A \cap B} = \bar{A} \cup \bar{B} \cdot \overline{A \cup B} = \bar{A} \cap \bar{B}$

Fuzzy

 $\mu_{\tilde{A}} : X \rightarrow [0, 1] \cdot \tilde{A} = \{(x, \mu(x)) \mid x \in X\}$ · Fuzzification: $x \mapsto \mu(x)$ Discrete: $\tilde{A} = \sum_{x_i \in X} \mu(x_i)/x_i$ · Continuous: $\tilde{A} = \int_X \mu(x)/x$ / = marker, not division · $\Sigma/\int$ = union of membership grades · membership $\neq$ probability

Top exam traps

1. Fuzzification = input · Defuzzification = output.
2. Membership grade is **not** probability.
3. Identity laws = **four**; empty = φ ; universal on Identity slide = X .
4. / in fuzzy notation is a **marker**, never division.
5. De Morgan bars cover the whole left-hand expression.
6. $A - B$ is not commutative; Cartesian pairs are ordered.

Glossary

Universe of discourse X/U · **Membership function** μ · **Fuzzy set** $\tilde{A}$ · **Grade** $\mu(x) \in [0, 1]$ · **Rule Base** IF–THEN expert rules · **Fuzzifier** crisp $\rightarrow$ fuzzy · **Inference Engine** match & fire · **Defuzzifier** fuzzy $\rightarrow$ crisp · **Cardinality** $|A|$ · φ empty set · **X** universal (Identity slide)